PRAGNA K N

Artificial Intelligence & Data Science Student
+91 7411968637 | pragna16kn@gmail.com | Yeshwanthpura, Bengaluru, Karnataka

OBJECTIVE

Motivated Artificial Intelligence and Data Science student with a strong foundation in Machine Learning, Deep Learning, and Computer Vision, complemented by hands-on Full Stack Development experience. Skilled in designing and deploying real-world AI applications, from intelligent detection systems to optimization tools, with a focus on translating research into practical, impactful solutions. Seeking opportunities to apply technical expertise to solve complex, real-world problems.

EDUCATION

Ramaiah Institute of Technology, Bengaluru *2023 – Present*
B.E. in Artificial Intelligence & Data Science | CGPA: 9.06

Milestone PU College *2021 – 2023*
PCMB | 95.6%

CKS English High School *2021*
SSLC | 92.3%

WORK EXPERIENCE

Full Stack Developer Intern — Parishrama Association *Nov 2025 – Jan 2026*

- Designed, developed, and maintained the official association website, building event update, gallery, and contact form modules.
- Implemented new features to improve user experience and ensure cross-device compatibility.
- Optimized website performance and coordinated with association members to deliver timely updates.

PROJECTS

RapidAid – Vision-Based Automated Accident Detection and Victim Condition Assessment

- Developed an AI-powered accident detection system using YOLOv8 and YOLOv8-Pose for real-time accident and victim analysis from video feeds.
- Implemented victim counting, pose estimation, and severity classification (Minor, Moderate, Critical) to support rapid emergency triage.
- Built structured emergency alert generation using Python, PyTorch, OpenCV, and Ultralytics YOLOv8.

Satellite Cloud Segmentation using Deep Learning

- Developed a cloud segmentation model using an EfficientNet-B0 U-Net architecture on the 38-Cloud satellite dataset.
- Implemented preprocessing, data augmentation, weighted sampling, and a combined Dice + BCE loss to improve cloud mask prediction accuracy.
- Built using Python, PyTorch, and OpenCV for end-to-end satellite image analysis.

Hybrid Quantum-Classical Optimization for Wind Farm Layout

- Formulated wind turbine placement as a QUBO optimization problem and solved it using the Quantum Approximate Optimization Algorithm (QAOA), achieving 13.7% higher Annual Energy Production over greedy search.
- Integrated wake effect and terrain elevation modeling for realistic energy estimation.
- Developed a Flask-based web application with Three.js 3D visualization to present optimization results.

CERTIFICATIONS

- Machine Learning — Coursera
- Data Science Specialization — Coursera
- HTML & CSS for Web Development — Udemy
- Machine Learning A-Z: AI, Python & R — Udemy
- Data Science and Machine Learning Bootcamp — Udemy

SKILLS

Technical Skills: Python, Java, Machine Learning, Deep Learning, Computer Vision, PyTorch, OpenCV, HTML, CSS, JavaScript, React, Flask, Power BI, Git

Soft Skills: Team Collaboration, Problem Solving, Communication, Adaptability

Languages: English, Kannada, Tamil, Hindi